Sprint Report 3

Whole-Body Model Predictive Control for Industrial Humanoids

Unit: EAT40006 – Engineering Technology Project B
Project: Whole-Body Model Predictive Control for Industrial Humanoids

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Submission Date: 14/10/2026

Contents

1. Acknowledgment of Country 2
2. Contribution summary (This Sprint) 2
3. SPrint plan 2
   - 3.1 Sprint Goal 3
   - 3.2 Sprint Objectives 3
   - 3.3 SPRINT BACKLOG 3
4. Final documentation deliverables 5
5. Sprint PRogress 6
   - 5.1 Existing Controller 6
   - 5.2 Disturbance recovery 7
   - 5.3 Walking State Machine 8
   - 5.4 Lateral Weight Shifting 8
   - 5.5 Footstep planning 9
   - 5.6 Swing-Foot Trajectory 11
   - 5.7 Inverse Kinematics 12
   - 5.8 Whole-body MPC 13
   - 5.9 CBF Safety Filter 14
   - 5.10 Single-Step Testing 15
   - 5.11 Multi-Step Testing 16
   - 5.12 Current Failure Analysis 16
   - 5.13 Reinforcement Learning Extension 17
   - 5.14 RL Training Development 19
   - 5.15 CBF cost issue 19
   - 5.16 Testing 19
6. Sprint review (Critical review of the product) 20
   - 6.1 Demonstration Session 20
   - 6.2 Professional Conduct 21
   - 6.3 Client Feedback 21
   - 6.4 Critical Analysis of Progress 21
7. Retrospect (Critical review of the process) 22
   - 7.1 Strengths 22
   - 7.2 Process Challenges 22
   - 7.3 Teamwork Review 23
   - 7.4 Professional Ethics 24
   - 7.5 Digital Ethics and Security 24
   - 7.6 AI Usage and Digital Ethics 24
   - 7.7 Robotics Safety Ethics 25
8. Lessons learned (Critical review of SPRINT one experience and future plan) 25

1. ACKNOWLEDGMENT OF COUNTRY

We acknowledge the Traditional Owners of the land on which this project was completed, the Wurundjeri Woi-wurrung People of the Kulin Nation, and pay our respects to Elders past and present.

We recognise their continuing connection to land, waters, culture and community. We extend that respect to all Aboriginal and Torres Strait Islander peoples.

2. CONTRIBUTION SUMMARY (THIS SPRINT)

All team members should complete the following table together. You can provide a yes/no response or a brief comment where appropriate.

	Contribution	Initiatives	Communication					Respect
Team Member	Finished tasks in time or on time with an acceptable quality	Self-assigned tasks and proactively found solutions to problems	Attended all team meetings	Attended all supervisor meetings	Attended all client meetings	Replied every time within an acceptable timeframe	Kept the team updated about the status	The student respected others and listened to different opinions
Rumeth	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Thinidu	No	Yes	No	Yes	No	Yes	Yes	Yes

During this sprint, the team consisted of two members and worked collaboratively on the humanoid control project. The major technical work was divided between implementation, testing, debugging, documentation and research.

Rumeth focused heavily on integrating and testing the control architecture, including the walking state machine, lateral weight shifting, swing-foot control, inverse kinematics, MPC integration, CBF safety filtering and the reinforcement-learning touchdown component.

Thinidu contributed to the project development, research, testing, documentation and discussions surrounding the reinforcement-learning and humanoid control approaches.

The team communicated through the agreed communication channels and maintained project files and implementation work within the project repository.

3. SPRINT PLAN

The Sprint 3 period covered Weeks 7–9. The primary objective of this sprint was to progress the project from a functioning standing-balance and disturbance-recovery framework towards a more complete humanoid stepping and walking framework. The sprint also focused on improving the quality of the

project implementation, testing the newly developed components, documenting the system architecture and preparing the project for demonstration and further research development.

3.1 SPRINT GOAL

The main goal of Sprint 3 was:

> Extend the existing whole-body MPC balance controller towards stepping by implementing weight shifting, swing-foot control, walking state management and touchdown handling, while improving the documentation and evaluation of the control framework. A secondary goal was to investigate reinforcement learning as a possible method for improving the difficult transition between swing, touchdown and double support.

3.2 SPRINT OBJECTIVES

The planned objectives were:

1. Review and improve the existing control architecture.
2. Implement a walking state machine.
3. Implement lateral weight shifting between the two feet.
4. Implement swing-foot trajectory generation.
5. Implement inverse kinematics for swing-leg control.
6. Integrate capture-point-based support checks.
7. Integrate the new walking components with the existing MPC controller.
8. Maintain the CBF as the final safety filtering layer.
9. Test single-step and multi-step behaviour.
10. Investigate the use of reinforcement learning around touchdown.
11. Develop training and evaluation infrastructure for the RL component.
12. Improve project documentation.
13. Prepare demonstrations and evidence of the current system.
14. Identify the remaining limitations and define the next sprint's technical objectives

3.3 SPRINT BACKLOG

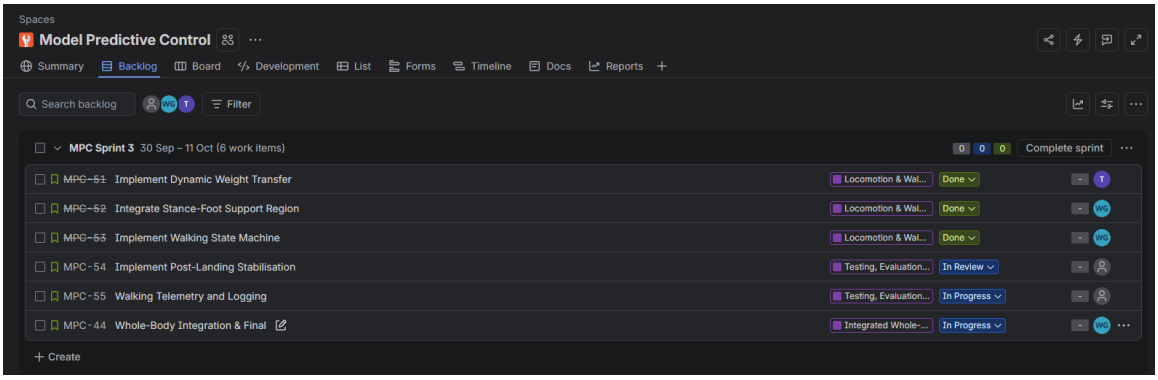


Figure 1 : Jira sprint backlog

Task	*Description*	*Priority*	*Expected Outcome*	*Status*
Review controller architecture	Review the existing MPC, PD and CBF architecture before extending it	High	Stable base architecture	Completed
Walking state machine	Implement standing, weight-shift, swing and double-support states	High	Automated stepping sequence	Completed
Lateral weight shifting	Transfer CoM onto the stance foot before swing	High	More stable swing initiation	Completed
Capture-point checks	Include CoM velocity and capture-point information in support readiness	High	More appropriate swing gating	Completed
Footstep planning	Generate alternating left/right footstep locations	High	Forward stepping targets	Completed
Swing trajectory	Generate smooth lift, forward movement and landing trajectory	High	Controlled swing foot	Completed
Leg IK	Convert swing-foot positions into joint targets	High	Swing-leg joint commands	Completed
MPC integration	Integrate walking references with whole-body MPC	High	Unified walking controller	Completed
CBF integration	Maintain safety filtering during walking	High	Safe torque filtering	Completed
Single-step testing	Evaluate the complete stepping sequence	High	Identify failure points	Completed / Experimental
Multi-step testing	Test repeated alternating steps	High	Evaluate continuous walking	Experimental
RL touchdown policy	Develop a residual RL controller for touchdown	Medium	Learned CoM correction	Implemented / Experimental
RL training	Develop touchdown-specific training process	Medium	Train candidate policies	In progress
RL evaluation	Compare RL against deterministic baseline	Medium	Quantitative comparison	In progress
Documentation	Update architecture, implementation and experimental documentation	High	Final documentation progress	In progress
Demonstration preparation	Prepare technical demonstration of completed functionality	High	Client/supervisor demonstration	In progress

Figure 2 : Sprint backlog subtast breakdown

4. FINAL DOCUMENTATION DELIVERABLES

At this stage, most of the teams have a fully functioning application. It is a good time to finalize all the final deliverables, including the technical documentation that goes with project implementations. Note that these technical user documentations can include a Usual Manual and a System Installation (Deployment) Manual, for the user and system admin to use, respectively. There can be other documentation needed specific to the project. There are no templates for these documents as they are system/project specific. Teams are advised to follow common sense to create these documents from the user and system admin perspectives for their system, to provide clear, concise and instructive information, so that the user and the system admin can follow them to use and install/deploy their system. Depending on the project, there can be other documentation required. Please discuss and finalize these with the client.

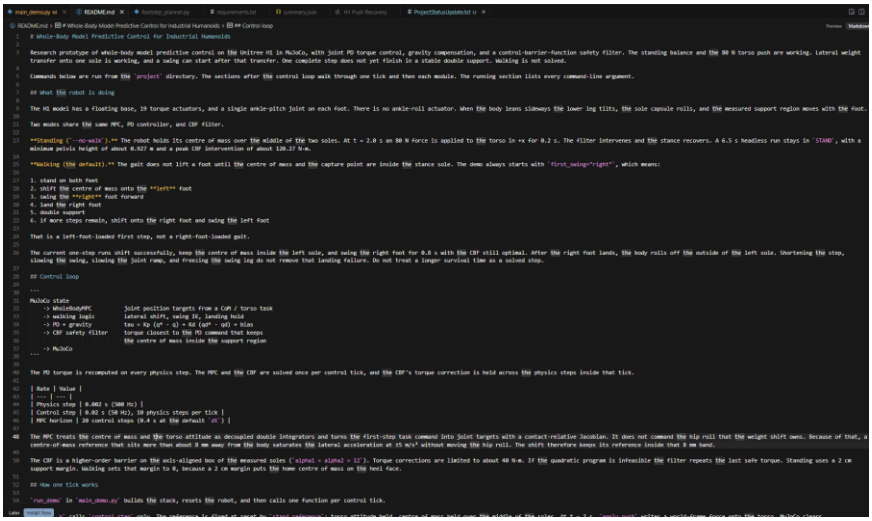


Figure 3 : User Manual and README document [APPENDIX A]



Figure 4 : Project status records [APPENDIX B]

5. SPRINT PROGRESS

The purpose of this sprint was to extend the previously developed standing and disturbance-recovery framework towards stepping and walking. The sprint made substantial progress in the control architecture.

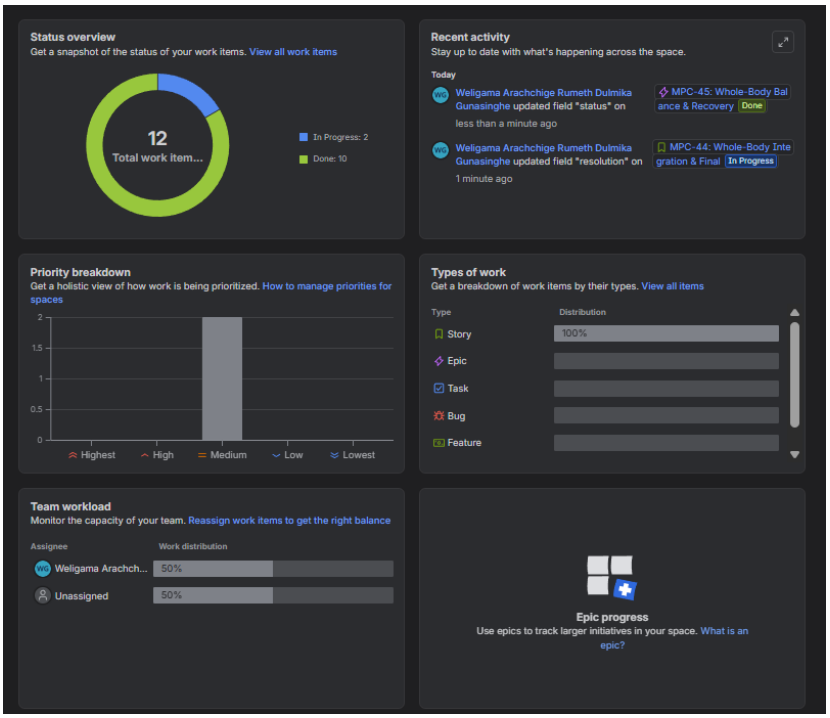


Figure 5 : Jira sprint progress

5.1 EXISTING CONTROLLER

The project initially provided a whole-body control framework for maintaining the stability of the Unitree H1 humanoid in MuJoCo. The original controller consisted of the MuJoCo simulation environment, robot state estimation, whole-body Model Predictive Control (MPC), joint-level PD control with gravity compensation, and a Control Barrier Function (CBF) safety filter.

The control process begins by obtaining the current robot state from MuJoCo, including joint positions, joint velocities, centre-of-mass (CoM) position and velocity, and torso orientation. The whole-body MPC then uses this state to generate desired joint positions for maintaining the desired CoM and torso behaviour. These desired positions are passed to the PD controller, which generates nominal joint torques. MuJoCo's qfrc_bias is also used to compensate for gravity and other bias forces.

The resulting nominal torque is then passed through the CBF safety filter. The CBF can modify the nominal torque when the estimated support constraints become critical. The final safe torque is then applied to the robot in MuJoCo.

The resulting control sequence is therefore:

MuJoCo state → Whole-Body MPC → PD + gravity compensation → nominal torque → CBF safety filter → safe torque → MuJoCo.

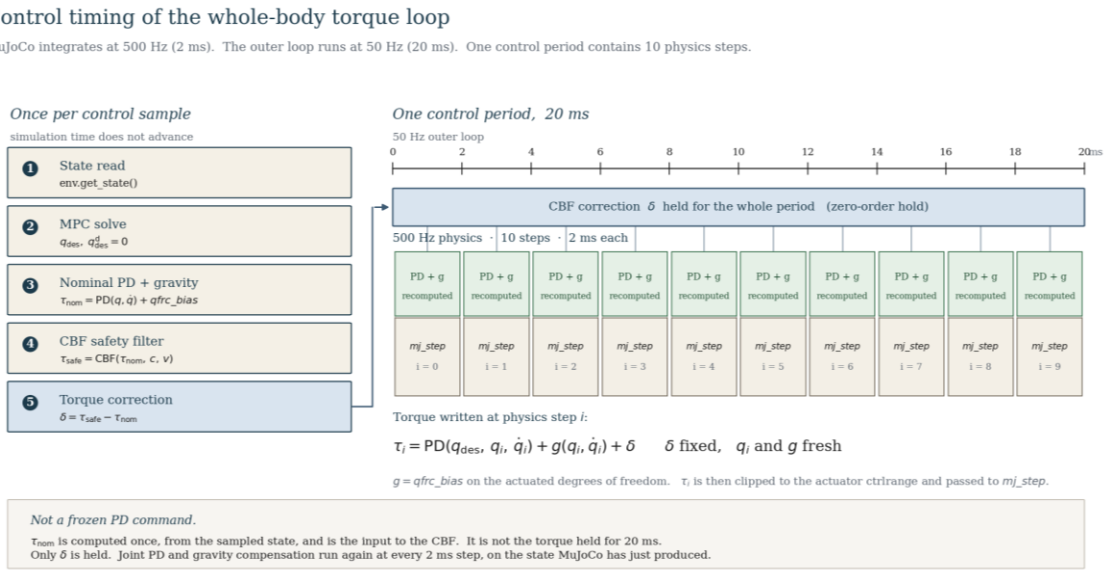


Figure 6 : The timing diagram

This provided the foundation for the stepping extension.

5.2 DISTURBANCE RECOVERY

Before extending the controller towards stepping, its ability to maintain balance under an external disturbance was evaluated. An 80 N force was applied in the forward direction to the torso for approximately 0.2 seconds.

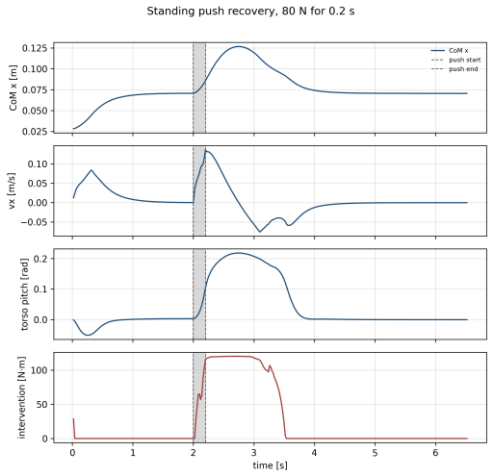


Figure 7 : Existing controller timing and disturbance-recovery behaviour

The experiment demonstrated that the robot was able to recover from the disturbance and remain upright. The previously recorded experiment produced a peak CBF intervention of approximately **120.27 N·m**, while the minimum robot height was approximately **0.927 m** and the final height was approximately **0.928 m**.

These results provided a baseline for the subsequent stepping work. In particular, the existing controller demonstrated that the robot could maintain standing balance and reject a significant external disturbance before additional walking behaviours were introduced.

5.3 WALKING STATE MACHINE

To extend the existing standing controller towards stepping, a walking state machine was implemented in walk_state_machine.py. The purpose of the state machine is to divide walking into distinct phases rather than attempting to control the complete stepping motion using a single controller behaviour.

The current state sequence alternates between weight transfer, swing and double-support phases. The main states implemented are:

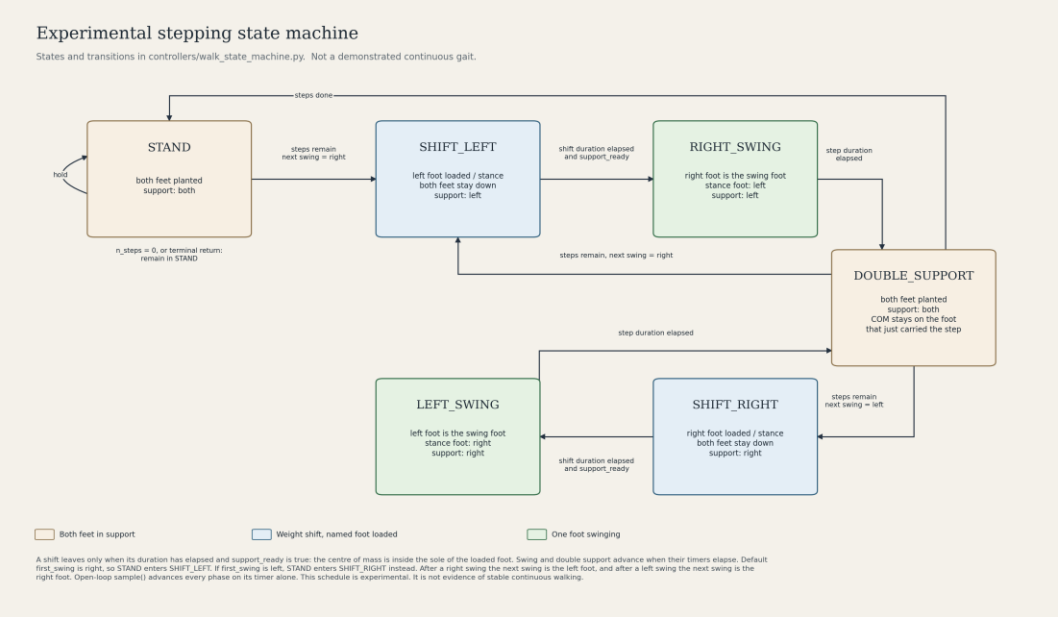


Figure 8 : walking state machine

For example, during the first step, the robot begins in STAND and then transfers its weight towards the left foot. Once the left foot has sufficient support, the right foot is released and enters the swing phase. After the right foot reaches the landing region, the controller enters the double-support phase before preparing for the next weight transfer.

The state machine therefore determines which phase of walking the robot is currently performing and allows the appropriate behaviour to be activated for that phase.

This separation is important because the control requirements are different during each stage. During weight transfer, the objective is to move the CoM towards the stance foot. During swing, the objective is to move the free foot towards its target while maintaining balance on the stance foot. After touchdown, the controller must recover stable double support before beginning the next step.

5.4 LATERAL WEIGHT SHIFTING

The lateral_shift.py module was developed to address one of the main problems introduced by stepping: the robot cannot safely lift one foot unless sufficient weight has first been transferred onto the other foot.

During standing, the H1 has both feet providing support. When one foot is lifted, the support region immediately becomes smaller because only the remaining stance foot can provide ground support. Therefore, the CoM must first be moved towards the stance foot. The lateral weight-shifting controller

uses the measured position of the stance foot and adjusts the robot's posture to move the CoM towards that foot. The controller does not simply command an arbitrary lateral joint movement. Instead, it considers the measured stance-foot location and progressively changes the posture until the required support condition is reached.

The weight-shifting process also controls the lateral velocity of the CoM. This is important because having the CoM positioned over the stance foot is not sufficient if the robot is still moving rapidly in the lateral direction. The controller therefore attempts to satisfy two conditions before releasing the swing foot:

- The CoM must be sufficiently close to the stance-foot support region.
- The lateral CoM velocity must be sufficiently small.

Once these conditions are satisfied, the walking state machine is allowed to progress into the swing phase.

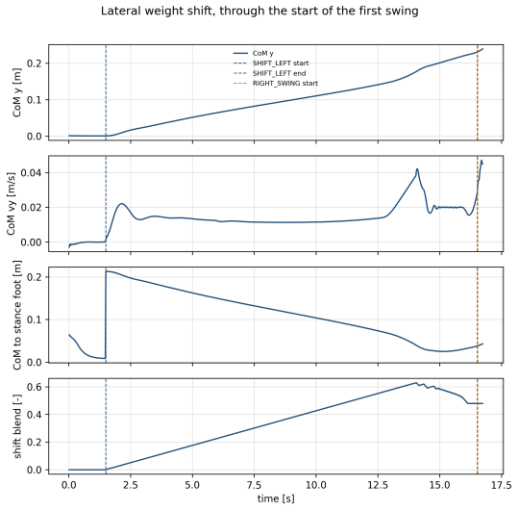


Figure 9 : weight shifting with steps

The figure shows the change in lateral CoM position and velocity during the weight-transfer phase. The reduction in the distance between the CoM and the stance foot demonstrates that the controller is actively moving the robot towards the supporting foot before swing begins.

This component was an important improvement over simply commanding the swing foot immediately because it provides a controlled transition from double support to single support.

5.5 FOOTSTEP PLANNING

The footstep_planner.py module was implemented to provide the high-level locations for the alternating footsteps.

The planner separates the decision of where the robot should place its foot from the lower-level controllers responsible for maintaining balance and moving the leg.

For the current experimental configuration, the planned forward progression is approximately:

- Right foot → 0.1 m

- Left foot → 0.2 m
- Right foot → 0.3 m
- Left foot → 0.4 m

These values represent the intended forward locations of successive footstep targets rather than completed physical footsteps.

The planner determines:

- which foot is the next swing foot;
- which foot remains as the stance foot;
- the forward landing location;
- the sequence of alternating footsteps;
- the target used by the swing-foot trajectory.

The planner therefore provides the high-level stepping targets, while the swing trajectory and inverse-kinematics modules are responsible for converting these targets into actual robot motion.

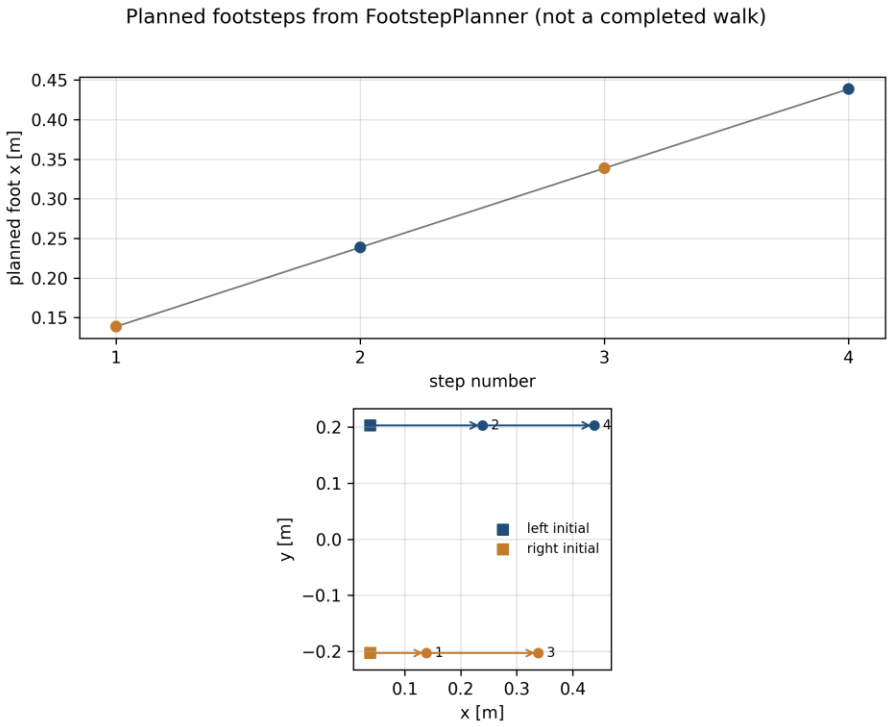


Figure 10 Experimental footstep plan

The figure shows the planned positions of the alternating footsteps. This demonstrates that the walking system has progressed beyond simply moving the CoM forward and now contains an explicit footstep-generation component.

An important limitation is that these targets represent the planned trajectory. Since the current robot does not yet complete the full four-step sequence stably, the figure should not be interpreted as evidence that all four footsteps were successfully achieved.

5.6 SWING-FOOT TRAJECTORY

Once the robot has transferred sufficient weight onto the stance foot, the swing foot can be lifted from the ground.
The swing-foot trajectory was implemented to generate a smooth path between the current foot position and the planned landing position. Rather than moving the foot directly along a straight line, the trajectory introduces a vertical lifting phase so that the foot clears the ground before moving forward.

1. The current trajectory can be understood as four stages:
2. Lift – the swing foot moves upwards from the ground.
3. Forward movement – the foot progresses towards the planned landing position.
4. Lowering – the foot returns towards the ground.
5. Touchdown – the foot reaches the landing region.

The current experimental swing height is approximately 0.05 m.
The trajectory is generated smoothly so that large instantaneous changes in foot position are avoided. This is important because abrupt swing-foot movement can produce large joint motions and make balance control more difficult.

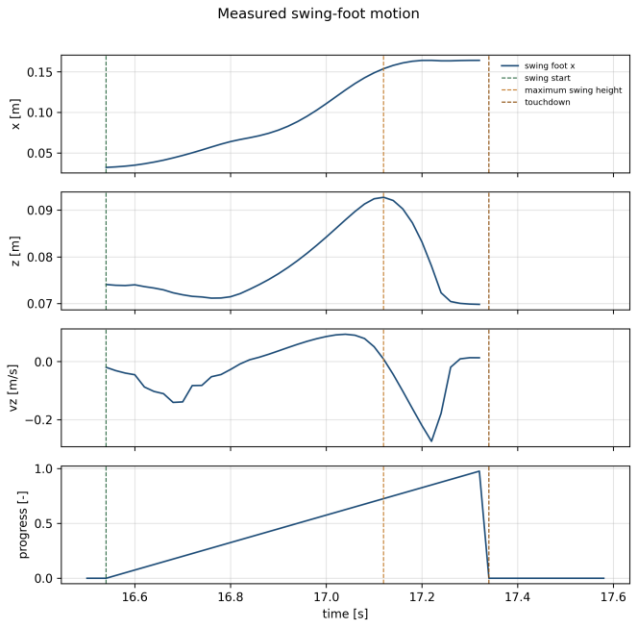


Figure 11 : Current swing tragectory graph

The current experimental swing height is approximately 0.05 m.

The swing trajectory prevents the foot from simply moving directly through the ground and provides smoother movement.

5.7 INVERSE KINEMATICS

The leg_ik.py module connects the Cartesian swing-foot trajectory to the robot's joint-space controller. The footstep planner and swing trajectory operate in Cartesian space because it is more intuitive to specify a desired foot position in terms of metres. However, the robot's actuators require joint-level commands.

The inverse-kinematics module therefore receives the desired swing-foot position and calculates the corresponding leg joint angles.

The current implementation uses a simplified two-link representation of the leg to calculate the main sagittal-plane joint angles:

- hip pitch;
- knee;
- ankle pitch.

The module also performs reachability and joint-limit checks before accepting the generated configuration.

This creates the following relationship:

> Footstep target → Cartesian swing trajectory → inverse kinematics → joint targets → PD controller → joint torque.

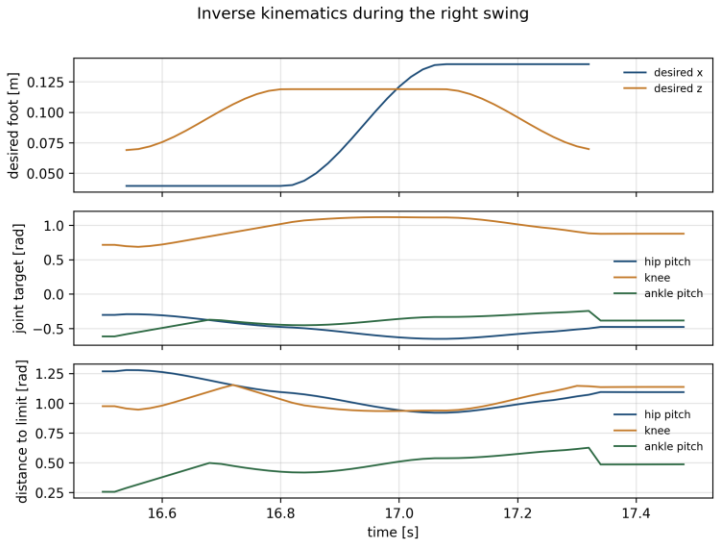


Figure 12 IK during the right swing

The figure shows the relationship between the desired swing-foot motion and the resulting hip-pitch, knee and ankle-pitch joint targets.

This demonstrates that the high-level footstep target is being translated into a physically meaningful joint-space trajectory rather than directly commanding arbitrary joint positions.

A limitation of the current implementation is that the IK model is simplified and primarily represents the leg's pitch-plane motion. It does not represent the complete whole-body inverse-kinematics problem.'

5.8 WHOLE-BODY MPC

The existing whole-body MPC was extended so that it could operate together with the new walking components.

The MPC maintains a six-dimensional task consisting of:

- CoM X;
- CoM Y;
- CoM Z;
- torso roll;
- torso pitch;
- torso yaw.

This allows the higher-level controller to simultaneously consider translational CoM behaviour and torso orientation.

The current MPC uses a prediction horizon of 20 steps with a timestep of 0.02 seconds, giving approximately 0.4 seconds of prediction.

At each control update, the MPC predicts how the robot's task variables can evolve and determines joint-position targets that move the robot towards the desired task state.

Contact-relative formulation

A particularly important modification for stepping was the use of a contact-relative CoM formulation.

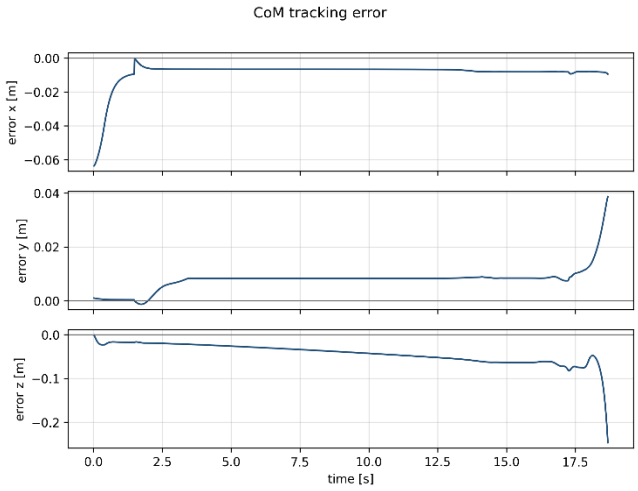


Figure 13 whole body MPC tracking and error

The figure demonstrates the relationship between the commanded task references and the resulting robot behaviour. The corresponding tracking-error plot shows how closely the CoM follows the desired reference during the experiment.

The H1 has a floating base, meaning that the base itself is not directly actuated in the same way as the robot's joints. If the controller considered only the global CoM Jacobian, the floating base could contribute to the mathematical solution without providing an actual actuated mechanism for achieving the required motion.

The controller therefore uses the relationship between the CoM Jacobian and the supporting-foot Jacobian. This makes the task more closely related to the motion that can actually be produced relative to the supporting contact.

5.9 CBF SAFETY FILTER

The CBF remains the final stage of the control architecture.

The complete control process is:

MPC → PD + gravity compensation → nominal torque → CBF → safe torque → MuJoCo.

The purpose of the CBF is not to replace the MPC or PD controller. Instead, it acts as a **safety filtering layer** between the nominal controller and the simulated robot.

The CBF represents the robot's current support region and formulates an optimisation problem to determine whether the nominal torque should be modified. If the nominal torque would move the estimated CoM towards an undesirable support condition, the filter can alter the torque while attempting to remain close to the original controller command.

This is particularly important during walking because the available support region changes when one foot is lifted.

The difference between nominal and safe torque provides a useful measurement of how strongly the safety layer is intervening.

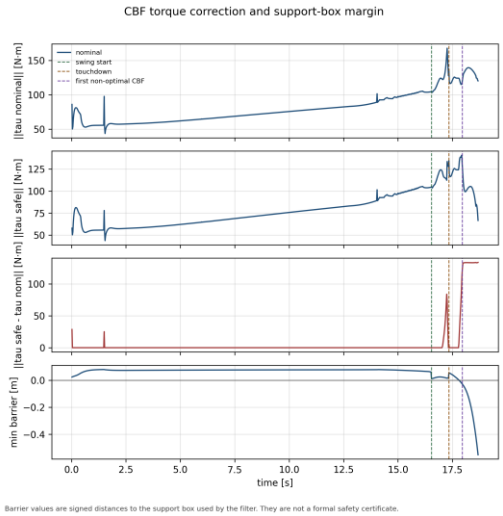


Figure 14 CBF_invention while moving

The figure should show:

- nominal torque;
- safe torque;
- CBF intervention magnitude;
- support/barrier margin.

During normal operation, the intervention should generally remain small. Larger interventions indicate that the safety layer is working harder to maintain the support constraints.

CBF limitation

An important limitation identified during the sprint is that the current implementation should not be described as a formal whole-body safety guarantee.
The relationship between applied torque and CoM acceleration is linearised around a nominal configuration. As the robot configuration changes significantly, particularly when a foot is lifted during walking, this approximation becomes less accurate.
Therefore, the CBF should be described as an experimental safety filter, rather than claiming that it mathematically guarantees that the humanoid cannot fall.

5.10 SINGLE-STEP TESTING

Single-step experiments were conducted to evaluate whether the individual components could operate together as a complete stepping sequence.

The experiment progresses through:

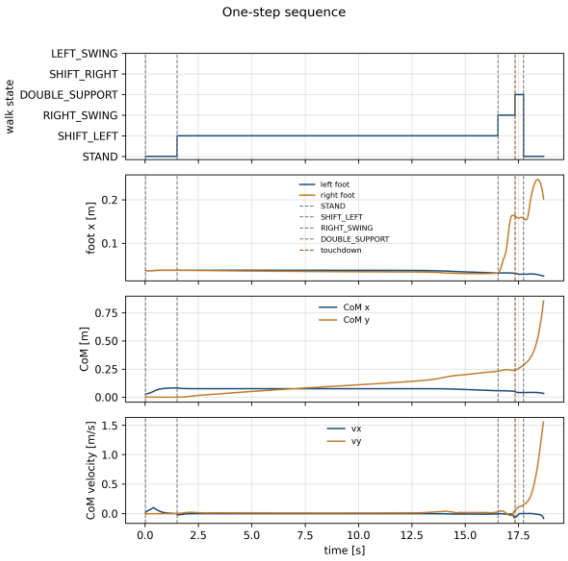
Standing → weight shift → right-foot swing → touchdown → double support.

The results show that the lateral weight-shifting controller successfully prepares the robot for swing and that the swing trajectory and inverse-kinematics components are capable of moving the right foot towards its landing target. The robot therefore progresses substantially further than the original standing-only controller.

However, the experiment also revealed that reaching touchdown is not equivalent to achieving a successful step. After touchdown, the robot develops significant lateral motion and eventually loses stability.

The one-step experiment therefore provides two important results:

- The stepping components can operate together sufficiently to reach touchdown.
- The complete landing and double-support recovery problem remains unresolved.



5.11 MULTI-STEP TESTING

Following the single-step experiment, the controller was tested using a multi-step schedule. The purpose of this experiment was to determine whether the walking state machine could repeatedly alternate between weight transfer, swing and double-support phases.The state machine successfully progresses into the intended sequence. However, the physical robot does not currently complete four stable steps.

The primary failure occurs during the transition:

late swing → touchdown → double support

In the four-step experiment, the first landing failure occurs before the robot can establish stable support for the next step. Consequently, the subsequent state transition occurs while the robot is already becoming unstable. This demonstrates that the main limitation is no longer simply generating a foot trajectory. The more difficult problem is maintaining dynamically recoverable whole-body motion during and immediately after touchdown.

5.12 CURRENT FAILURE ANALYSIS

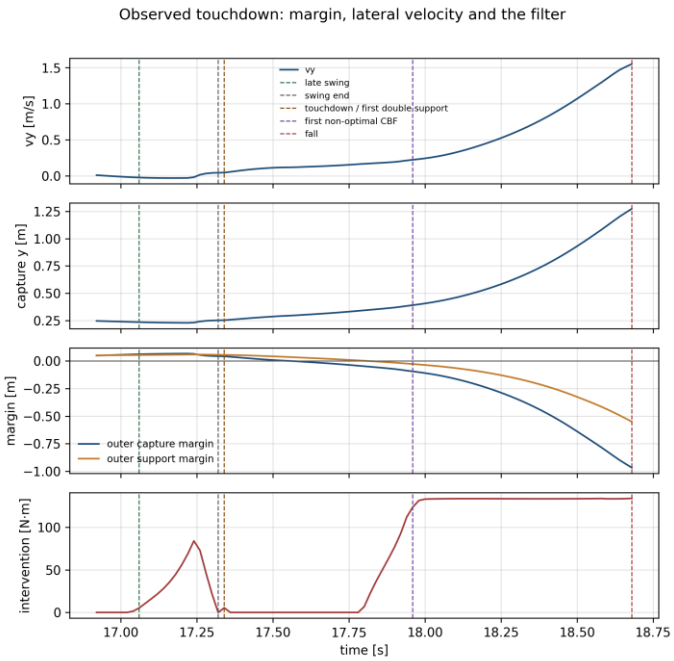
The current experiments indicate that the main unresolved problem is lateral stability during touchdown and double-support recovery. An important observation from the experiments is that geometric CoM position alone does not fully describe the robot's stability. Near the end of the right-foot swing, the CoM can still be located inside the available support region. However, the robot may already have developed lateral velocity.

One recorded state near the end of the swing was approximately:

- CoM ≈ (0.050, 0.240) m;
- CoM velocity ≈ (-0.070, +0.042) m/s.

At this point, the geometric support margin can still appear acceptable. However, after touchdown, the lateral velocity increases substantially and the robot eventually falls.The one-step experiment recorded a peak lateral velocity of approximately 1.55 m/s before the robot terminated due to the fall.

This observation suggests that the controller needs to consider not only where the CoM is, but also how the CoM is moving when touchdown occurs. This is why capture-point information is used in the current controller. The capture point incorporates both CoM position and velocity and therefore provides additional information about whether the current dynamic state is recoverable within the available support region. The current evidence does not establish that a single variable is solely responsible for the failure. Instead, it identifies the late-swing/touchdown transition as the main area requiring further investigation.



Observed traces from the deterministic one-step run. They do not by themselves prove a single cause.

Although the support margin can still appear acceptable, the robot subsequently develops a much larger outward velocity after touchdown.

This indicates that simply checking geometric CoM position is not sufficient. The robot must arrive at touchdown with a dynamically recoverable state.

5.13 REINFORCEMENT LEARNING EXTENSION

Because the main failure occurs during a relatively short and clearly identifiable portion of the walking cycle, a touchdown-specific reinforcement-learning controller was implemented as an experimental extension.Rather than using reinforcement learning to directly control all robot torques, the RL policy operates at a higher level.

The current architecture is:

> RL policy → small CoM reference residual → Whole-Body MPC → PD + gravity compensation → CBF → robot.

This design was selected so that the RL policy does not bypass the existing model-based controller or the safety layer. The RL action is a two-dimensional CoM reference residual:

- ΔCoM X;
- ΔCoM Y.

The residual is limited to approximately ±0.02 m. This means that the RL controller can make only a small adjustment to the existing reference rather than completely changing the robot's desired motion.

RL observation

The touchdown policy receives information describing the robot's dynamic state, including:

- CoM position;

- CoM velocity;
- capture point;
- capture-point error;
- support margins;
- pelvis orientation;
- stance-leg posture;
- swing-foot position;
- swing-foot velocity;
- swing progress;
- time since RL activation;
- CBF intervention.

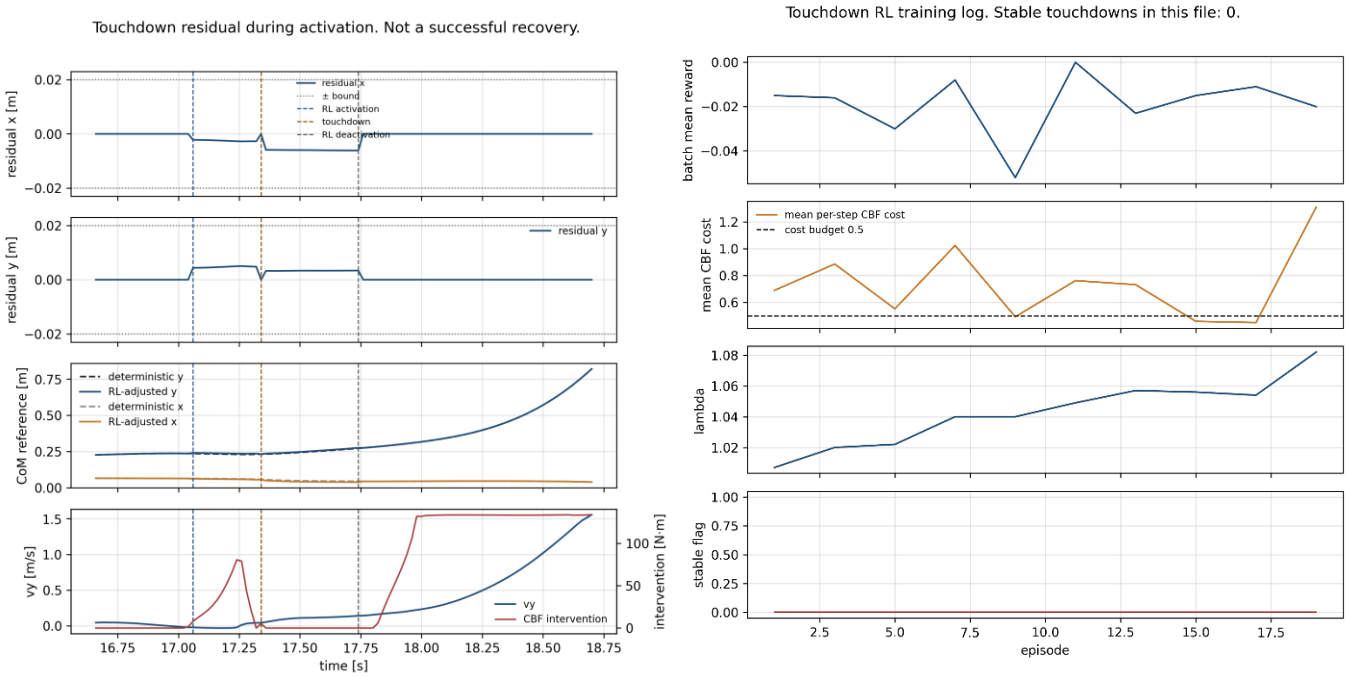


Figure 15 : Touchdown RL training and residual

The policy is activated primarily during the late-swing and touchdown portion of the walking cycle, where the deterministic controller currently has difficulty maintaining lateral stability.

Figure 17 – Touchdown RL residual and control response

The figure should show the RL residual over time together with the resulting CoM reference and relevant stability variables.

The purpose of this experiment is not to replace the existing controller with RL. Instead, RL is being investigated as a mechanism for learning a small corrective adjustment to the model-based reference specifically around the difficult touchdown phase.

5.14 RL TRAINING DEVELOPMENT

I would add this section. Your current report jumps from implementing RL straight to the next section, but your actual development work here is important.A dedicated touchdown training framework was developed rather than training the complete walking behaviour from the beginning.The deterministic walking controller first progresses the robot towards the late-swing stage. The resulting robot state is then captured and used as the starting point for the touchdown learning problem.

This reduces the size of the learning problem because the RL policy does not need to learn standing, weight shifting, swing-foot movement and walking from the beginning. Instead, it focuses specifically on the transition that has already been identified as problematic.

During training, the RL policy is evaluated using both performance and safety-related information. The reward considers factors including support margin, capture behaviour, velocity, posture and touchdown/stabilisation behaviour. CBF intervention is also incorporated into the cost used by the constrained learning formulation.

5.15 CBF COST ISSUE

During development, an inconsistency was identified in the way the CBF cost was being used by the RL constraint. The original implementation accumulated CBF cost across an entire episode but compared that accumulated value with a budget that was intended to represent a mean per-step cost. Since an episode contains many control steps, this caused the Lagrange multiplier to increase excessively and interfered with the learning process.

The formulation was subsequently corrected so that the constraint uses the mean per-step CBF cost. Following the correction, the Lagrange multiplier remained within a more appropriate operating range rather than immediately approaching its maximum. However, the corrected training process has not yet produced evidence that the RL controller improves touchdown stability.

The current 20-episode training/evaluation result produced no stable landings. Therefore, RL should currently be considered an experimental extension under development, rather than a completed solution to the walking problem.

5.16 TESTING

I would also add a proper testing subsection. Two dedicated test modules were implemented:

- tests/test_lateral_shift.py
- tests/test_touchdown.py

The lateral-shifting test suite currently passes 6 tests, while the touchdown test suite also passes 6 tests.

These tests are intended to verify the behaviour of individual software components and therefore provide confidence that the implementation is functioning according to the expected software-level conditions. However, component-level testing is different from full-system robotics validation.

Passing the unit tests does not demonstrate that the humanoid can successfully complete a step or walk continuously. The experimental results demonstrate that the individual modules can operate together

sufficiently to reach touchdown, while the full-system experiment continues to expose a stability problem during touchdown and double support.

This distinction is important because a robotics system can contain correctly implemented individual components while still exhibiting unstable emergent behaviour when those components interact dynamically.

6. SPRINT REVIEW (CRITICAL REVIEW OF THE PRODUCT)

The sprint review focused on evaluating the progress of the humanoid control framework against the sprint objectives.

6.1 DEMONSTRATION SESSION

The demonstration was structured to show the project incrementally rather than attempting to demonstrate the entire system at once.

The planned demonstration sequence was:

Demonstration 1 — Standing Controller

The Unitree H1 is started in the standing configuration. The demonstration shows the operation of:

- whole-body MPC;
- PD control;
- gravity compensation;
- CBF safety filtering.

Demonstration 2 — Disturbance Recovery

An 80 N external force is applied to the torso. The robot is observed while it responds to the disturbance and attempts to return to a stable configuration. This demonstrates the most mature part of the control framework.

Demonstration 3 — Weight Transfer

The walking controller transfers the CoM towards the stance foot. This demonstrates the newly implemented lateral weight-shift controller.

Demonstration 4 — Swing Foot

The swing foot is lifted, moved forward and lowered according to the generated trajectory. This demonstrates the footstep planner, trajectory generator and inverse kinematics.

Demonstration 5 — Touchdown

The robot attempts to complete the transition into double support. This demonstrates the current limitation of the system and provides the basis for further research.

Demonstration 6 — RL Touchdown

The experimental RL architecture is explained as a possible method for improving the touchdown transition.

6.2 PROFESSIONAL CONDUCT

Professional communication was maintained during project meetings and discussions.

The team attempted to communicate project status, technical progress and difficulties clearly with the supervisor and client.

When technical difficulties prevented expected progress, these issues were communicated rather than being hidden.

The team also maintained professional behaviour during technical discussions by:

- listening to different technical opinions;
- discussing alternative approaches;
- documenting decisions;
- communicating changes to the implementation;
- acknowledging limitations in the current system;
- avoiding unsupported claims about project performance.

6.3 CLIENT FEEDBACK

During the client demonstration, the team presented the current control architecture, standing disturbance recovery, stepping components and the current touchdown limitation. The client was informed that the system has successfully demonstrated standing and disturbance recovery but that continuous stable walking has not yet been achieved.

The client commented that we should focus on the Reinforcement learning layer and overcoming the current touchdown failure and discuss further action with the colleague that was referred by the client.

6.4 CRITICAL ANALYSIS OF PROGRESS

The sprint achieved most of the software implementation objectives.

The major achievement was extending the original standing controller into a more comprehensive stepping architecture.

The following components were successfully implemented:

- walking state machine;
- lateral weight shifting;
- capture-point checks;
- footstep planning;
- swing-foot trajectory;
- inverse kinematics;
- MPC integration;
- CBF integration;
- touchdown detection;
- RL touchdown infrastructure;
- automated component tests.

However, the major planned behavioural objective, reliable walking has not yet been achieved. The main reason is that the transition from swing to touchdown introduces a more difficult dynamic control problem than the original standing task. The current system can successfully reach the touchdown stage but can develop excessive lateral velocity during late swing and become unstable after touchdown. This has changed the focus of the project from simply implementing more walking functionality to understanding and solving the touchdown stability problem.

7. RETROSPECT (CRITICAL REVIEW OF THE PROCESS)

7.1 STRENGTHS

- **Clear separation of controller responsibilities**

The project architecture became more modular during the sprint. Different components now have clearly defined responsibilities:

This made debugging easier because individual behaviours could be investigated separately.

- **Incremental development**

Rather than attempting to implement walking immediately, the team developed the system progressively. This allowed problems to be isolated more effectively.

- **Evidence-based debugging**

The team increasingly relied on logged quantities rather than only observing the robot visually. Important measurements included:

- CoM;
- CoM velocity;
- capture point;
- support margin;
- CBF intervention;
- torso orientation;
- joint positions.

This helped identify the current lateral instability problem.

7.2 PROCESS CHALLENGES

Challenge 1 - Small Team Size

The project was being developed by only two team members. This meant that:

- technical implementation had to be shared;
- documentation competed with development time;
- debugging responsibilities were concentrated among fewer people;
- absence or reduced availability of one member had a significant effect on progress.
- **Response :**The team reduced unnecessary parallel work and prioritised the highest-impact technical tasks.

Challenge 2 - Complex Integration

The individual controllers could work independently, but integrating them introduced new interactions. For example:

- the MPC can influence hip roll;
- the weight-shift controller also needs hip roll;
- swing-foot control changes the support configuration;
- the CBF uses the support region;
- changing contact configuration changes the robot dynamics.
- **Response :** The architecture was separated into clearly defined control responsibilities.

The weight-shift controller temporarily owns specific posture variables where necessary, while the MPC continues to handle whole-body motion.

Challenge 3 — Walking Instability

The largest technical bottleneck was the failure during touchdown. The robot could successfully reach the landing stage but develop excessive lateral motion.

- **Response**

The team:

1. Added capture-point checks.
2. Added lateral weight shifting.
3. Adjusted hip-roll behaviour.
4. Adjusted ankle-pitch behaviour.
5. Added swing-target ramping.
6. Tested different step lengths.
7. Tested different swing durations.
8. Tested different touchdown references.
9. Developed touchdown diagnostics.
10. Began investigating an RL residual controller.

The problem has not yet been completely solved, but the failure mode is now better characterised.

Challenge 4 — RL Training

The RL system initially had a cost-scaling issue. The CBF cost was accumulated over the episode while the intended constraint was based on per-step behaviour.

- **Response :** The cost calculation was changed to use a mean per-step CBF cost. This improved the scaling of the Lagrangian constraint. However, further training and evaluation are still required.

7.3 TEAMWORK REVIEW

The team maintained a collaborative approach to development.

Technical decisions were discussed before major changes were made, and progress was communicated between members.

The team also recognised that a two-person project requires clear ownership of tasks and frequent status updates because a delay affecting one member can significantly affect the entire project.

7.4 PROFESSIONAL ETHICS

The project involves a humanoid robot controller intended for research and simulation. The team therefore considered the ethical implications of claiming that the controller is safe or reliable. An important ethical principle adopted during the sprint was: **Do not claim functionality that has not been demonstrated.**

For example, the current system should not be described as a successful walking controller because continuous stable walking has not yet been demonstrated.

Similarly, the CBF should not be described as providing a formal guarantee of whole-body robot safety because the current implementation has modelling and contact-configuration limitations. bThis distinction is important because exaggerated claims about robot safety could be misleading if the controller were later applied to a physical robot.

7.5 DIGITAL ETHICS AND SECURITY

The project primarily operates in a simulation environment and does not intentionally collect personal information. The main digital-ethics and security considerations are therefore associated with:

- source-code security;
- dependency management;
- repository access;
- experiment reproducibility;
- responsible use of AI-assisted development;
- integrity of experimental results;
- protection of project files;
- responsible reporting of system limitations.

No personal information is required for the humanoid simulation. Where privacy legislation such as the Australian Privacy Principles is considered, its relevance to this project is limited because the project does not intentionally process personal information. Nevertheless, the team will avoid introducing unnecessary personal data into project repositories, logs or datasets.

7.6 AI USAGE AND DIGITAL ETHICS

AI tools may be used during development for activities such as:

- debugging assistance;
- explaining technical concepts;
- generating suggestions;
- improving documentation structure.

However, AI-generated material must be reviewed by the team before inclusion in the project. The team remains responsible for:

- verifying technical claims;

- testing generated code;
- understanding submitted material;
- correctly acknowledging AI use where required;
- ensuring that AI does not replace genuine engineering analysis.

Experimental results will be based on actual simulation runs rather than AI-generated claims.

7.7 ROBOTICS SAFETY ETHICS

Although the current project is simulated, the team recognises that the eventual application is a physical humanoid robot.

Therefore:

- safety constraints should not be deliberately disabled to produce better-looking demonstrations;
- torque limits should be respected;
- unstable behaviours should be reported rather than hidden;
- experimental changes should be tested in simulation before physical deployment;
- RL should not be allowed to bypass the established safety layer.

The current architecture intentionally keeps the CBF after the RL/MPC/PD stages.

8. LESSONS LEARNED (CRITICAL REVIEW OF SPRINT ONE EXPERIENCE AND FUTURE PLAN)

Implementing walking is significantly harder than implementing standing balance

- The standing controller can maintain a relatively fixed support configuration.
- Walking changes the support configuration and introduces hybrid transitions between:
- double support;
- single support;
- touchdown;
- double support again.
- This means that a controller that works well for standing cannot automatically be expected to produce stable walking.
- **Action :** Future development will focus more specifically on contact transitions rather than only improving the standing controller.

Geometric stability is not enough

- One of the major lessons from the experiments was that having the CoM inside the support region does not necessarily mean that the robot is dynamically stable.
- CoM velocity and capture point must also be considered.
- **Action :** Capture-point and velocity-based criteria will continue to be used when determining whether the robot is ready for touchdown and the next step.

Safety filtering should not replace good control

- The CBF can modify a nominal command, but it should not be treated as a mechanism that automatically fixes every unstable state.

- **Action :** The nominal controller will be improved so that the robot reaches touchdown in a more recoverable state before relying on the CBF.

RL needs a clearly defined problem

- Training RL over the entire walking system would introduce a very large search space.
- The touchdown-specific formulation provides a much smaller and more targeted problem.
- **Action :** Continue evaluating RL specifically around the late-swing/touchdown transition before considering a larger RL controller.